

Recall:

Let A be a matrix with a singular value decomposition

$$A = U\Sigma V^T$$

If

$$U = [\mathbf{u}_1 \ \dots \ \mathbf{u}_m] \quad V = [\mathbf{v}_1 \ \dots \ \mathbf{v}_n]$$

and $\sigma_1, \dots, \sigma_r$ are singular values of A then then

$$A = \sigma_1(\mathbf{u}_1\mathbf{v}_1^T) + \sigma_2(\mathbf{u}_2\mathbf{v}_2^T) + \dots + \sigma_r(\mathbf{u}_r\mathbf{v}_r^T)$$

Example: Movie ratings:

	Matrix	Amelie	Alien	Casablanca	Interstellar
user_1	5	0	5	0	4
user_2	5	0	3	0	5
user_3	0	5	0	5	1
user_4	1	5	0	4	0
user_5	4	0	4	0	3
user_6	0	5	0	4	0
user_7	3	0	3	0	2